

Experiment no-21

21. Create a python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

PROGRAM:

```
import nltk
from nltk import word_tokenize, pos_tag
from nltk.chunk import RegexpParser

# Download required resources (Run only once)
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')

sentence = input("Enter a sentence: ")

words = word_tokenize(sentence)
tags = pos_tag(words)

grammar = "NP: {<DT>?<NN.*>+}"

parser = RegexpParser(grammar)
tree = parser.parse(tags)

print("Noun Phrases:")
for subtree in tree.subtrees():
    if subtree.label() == "NP":
        print(" ".join(word for word, tag in subtree.leaves()))
```

OUTPUT:

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
Enter a sentence: The intelligent student solved the difficult problem
Noun Phrases:
The intelligent student
the difficult problem
```